Raniya N

Data Science

Pandas Data Integration and Transformation

MERGING DATA USING PANDAS

Basic Merging :

In [4]:
```
import pandas as pd

df1 = pd.DataFrame({
    "id": [1, 2, 3],
    "name": ["Asha", "Binu", "Cathy"]
})

df2 = pd.DataFrame({
    "id": [1, 2, 4],
    "marks": [85, 90, 75]
})

merged_df = pd.merge(df1, df2, on="id")
print(merged_df)
```

```
   id  name  marks
0   1  Asha     85
1   2  Binu     90
```

Types of Joins :

Inner Join (default)

Only common rows:

In [7]:
```
pd.merge(df1, df2, on="id", how="inner")
```

Out[7]:

	id	name	marks
0	1	Asha	85
1	2	Binu	90

Left Join

All rows from left DataFrame:

In [9]:

```
pd.merge(df1, df2, on="id", how="left")
```

Out[9]:

	id	name	marks
0	1	Asha	85.0
1	2	Binu	90.0
2	3	Cathy	NaN

Right Join

All rows from right DataFrame:

In [11]:

```
pd.merge(df1, df2, on="id", how="right")
```

Out[11]:

	id	name	marks
0	1	Asha	85
1	2	Binu	90
2	4	NaN	75

Outer Join

All rows from both:

In [13]:

```
pd.merge(df1, df2, on="id", how="outer")
```

Out[13]:

	id	name	marks
0	1	Asha	85.0
1	2	Binu	90.0
2	3	Cathy	NaN
3	4	NaN	75.0

Merge on Different Column Names

Merge on Different Column Names means combining two DataFrames using matching columns that have different names in each DataFrame.

Example:

First DataFrame → id Second DataFrame → student_id

Even though names are different, they contain related data, so pandas merges them using left_on and right_on.

In [16]:

```
import pandas as pd

df1 = pd.DataFrame({
    "id": [1, 2, 3],
    "name": ["Asha", "Binu", "Cathy"]
})

df2 = pd.DataFrame({
    "student_id": [1, 2, 4],
    "marks": [85, 90, 75]
})

merged_df = pd.merge(
    df1,
    df2,
    left_on="id",
    right_on="student_id"
)

merged_df
```

Out[16]:

	id	name	student_id	marks
0	1	Asha	1	85
1	2	Binu	2	90

Concatenation

In [18]:
```python
pd.concat([df1, df2], axis=0) # row-wise
```

Out[18]:

	id	name	student_id	marks
0	1.0	Asha	NaN	NaN
1	2.0	Binu	NaN	NaN
2	3.0	Cathy	NaN	NaN
0	NaN	NaN	1.0	85.0
1	NaN	NaN	2.0	90.0
2	NaN	NaN	4.0	75.0

In [37]:
```python
pd.concat([df1, df2], axis=1) # column-wise
```

Out[37]:

	id	name	student_id	marks
0	1	Asha	1	85
1	2	Binu	2	90
2	3	Cathy	4	75

Apply

In [21]:
```python
import pandas as pd

marks = pd.Series([45, 60, 75, 90])

def grade(x):
    if x >= 50:
        return "Pass"
    else:
        return "Fail"

print(marks.apply(grade))
```

```
0    Fail
1    Pass
2    Pass
3    Pass
dtype: str
```

Apply Using Lambda Function

In [23]:
```python
import pandas as pd

marks = pd.Series([45, 60, 75, 90])
```

```python
result = marks.apply(lambda x: x + 5)

print(result)
```

```
0    50
1    65
2    80
3    95
dtype: int64
```

apply() on DataFrame Columns

In [25]:

```python
import pandas as pd

df = pd.DataFrame({
    "Math": [80, 70, 90],
    "Science": [85, 60, 95]
})

print(df.apply(max))
```

```
Math       90
Science    95
dtype: int64
```